

SQL PROJECT

Mobile Phone Store Management – SQL Project

3 Tables : Customers,Phones,Sales

1.Customers Table:

```
CREATE TABLE customers (  
  
    customer_id INT PRIMARY KEY,  
  
    first_name VARCHAR(50),  
  
    last_name VARCHAR(50),  
  
    city VARCHAR(50),  
  
    phone_number VARCHAR(15),  
  
    registration_date DATE  
  
);
```

	customer_id	first_name	last_name	city	phone_number	registration_date
▶	1	Arun	Kumar	Kochi	9876543210	2025-01-05
	2	Anjali	Nair	Thiruvananthapuram	9876501234	2025-01-12
	3	Rahul	Menon	Kozhikode	9123456780	2025-02-01
	4	Neha	Thomas	Kochi	9012345678	2025-02-15
	5	Vishnu	Raj	Thrissur	9345678901	2025-03-03
	6	Akhil	Das	Kollam	8899776655	2025-03-18
	7	Meera	Suresh	Kochi	9988776655	2025-04-02
	8	Adithya	Krishnan	Kannur	8765432109	2025-04-19
	9	Fathima	Salim	Malappuram	9090909090	2025-05-10
	10	Nikhil	Paul	Kottayam	8080808080	2025-05-22
	11	Sneha	Joseph	Kochi	7999888777	2025-06-06
	12	Manu	Babu	Alappuzha	7888999000	2025-06-20
	13	Devika	Mohan	Palakkad	7777888899	2025-07-04
	14	Sanjay	Varma	Kochi	7666777888	2025-07-15
	15	Aparna	Vijay	Ernakulam	7555666777	2025-08-01

2.Phones Table:

```
CREATE TABLE phones ( phone_id INT PRIMARY KEY, brand  
VARCHAR(50), model VARCHAR(100), price DECIMAL(10,2), ram_gb  
INT, storage_gb INT );
```

	phone_id	brand	model	price	ram_gb	storage_gb
	1	Apple	iPhone 15	69900	6	128
	2	Apple	iPhone 16	79900	8	128
	3	Samsung	Galaxy S24	74999	8	256
	4	Samsung	Galaxy A55	39999	8	128
	5	OnePlus	12R	42999	8	256
	6	OnePlus	Nord CE 4	24999	8	128
	7	Google	Pixel 8	54999	8	128
	8	Xiaomi	Redmi Note 13 Pro	25999	8	256
	9	Realme	GT 6T	30999	12	256
	10	Motorola	Edge 50 Fusion	22999	8	128
	11	Vivo	V30	33999	8	256
	12	Vivo	Reno 12	32999	8	256
	13	Nothing	Phone 2a	23999	8	128
	14	Samsung	Galaxy S23 FE	49999	8	128
▶	15	Apple	iPhone 14	59900	6	128

3.Sales Table:

```
CREATE TABLE sales ( sale_id INT PRIMARY KEY, customer_id INT,  
phone_id INT, sale_date DATE, quantity INT, total_amount  
DECIMAL(10,2), FOREIGN KEY (customer_id) REFERENCES  
customers(customer_id), FOREIGN KEY (phone_id) REFERENCES  
phones(phone_id) );
```

	sale_id	customer_id	phone_id	sale_date	quantity	total_amount
▶	1	1	2	2025-02-03	1	79900
	2	2	4	2025-02-08	1	39999
	3	3	3	2025-02-18	1	74999
	4	4	6	2025-02-25	1	24999
	5	5	8	2025-03-10	2	51998
	6	6	10	2025-03-22	1	22999
	7	7	1	2025-04-05	1	69900
	8	8	9	2025-04-25	1	30999
	9	9	11	2025-05-14	1	33999
	10	10	5	2025-05-30	1	42999
	11	1	7	2025-06-12	1	54999
	12	11	13	2025-06-28	2	47998
	13	12	12	2025-07-08	1	32999
	14	13	14	2025-07-20	1	49999
	15	14	2	2025-08-05	1	79900
	16	15	15	2025-08-12	1	59900
	17	3	4	2025-08-20	1	39999

Question based on the mobile data sheet

1. List all phones with price above ₹30000.

Select * from phones where price > 30000;

Result Grid

	phone_id	brand	model	price	ram_gb	storage_gb
▶	1	Apple	iPhone 15	69900	6	128
	2	Apple	iPhone 16	79900	8	128
	3	Samsung	Galaxy S24	74999	8	256
	4	Samsung	Galaxy A55	39999	8	128
	5	OnePlus	12R	42999	8	256
	7	Google	Pixel 8	54999	8	128
	9	Realme	GT 6T	30999	12	256
	11	Vivo	V30	33999	8	256
	12	Oppo	Reno 12	32999	8	256
	14	Samsung	Galaxy S23 FE	49999	8	128
	15	Apple	iPhone 14	59900	6	128

2. Find all phones with 8 GB Ram.

Select * from phones where ram_gb = 8;

Result Grid

	phone_id	brand	model	price	ram_gb	storage_gb
▶	2	Apple	iPhone 16	79900	8	128
	3	Samsung	Galaxy S24	74999	8	256
	4	Samsung	Galaxy A55	39999	8	128
	5	OnePlus	12R	42999	8	256
	6	OnePlus	Nord CE 4	24999	8	128
	7	Google	Pixel 8	54999	8	128
	8	Xiaomi	Redmi Note 13 Pro	25999	8	256
	10	Motorola	Edge 50 Fusion	22999	8	128
	11	Vivo	V30	33999	8	256
	12	Oppo	Reno 12	32999	8	256
	13	Nothing	Phone 2a	23999	8	128
	14	Samsung	Galaxy S23 FE	49999	8	128

3. Find all phones from the Apple brand.

Select * from phones where brand = 'Apple';

Result Grid

	phone_id	brand	model	price	ram_gb	storage_gb
▶	1	Apple	iPhone 15	69900	6	128
	2	Apple	iPhone 16	79900	8	128
	15	Apple	iPhone 14	59900	6	128

4. Find all sales where quantity is greater than 1.

Select * from sales where quantity > 1;

Result Grid

	sale_id	customer_id	phone_id	sale_date	quantity	total_amount
▶	5	5	8	2025-03-10	2	51998
	12	11	13	2025-06-28	2	47998
	18	4	6	2025-08-23	2	49998

5. Find all sales above ₹50,000.

Select * from sales where total_amount > 50000;

Result Grid

	sale_id	customer_id	phone_id	sale_date	quantity	total_amount
▶	1	1	2	2025-02-03	1	79900
	3	3	3	2025-02-18	1	74999
	5	5	8	2025-03-10	2	51998
	7	7	1	2025-04-05	1	69900
	11	1	7	2025-06-12	1	54999
	15	14	2	2025-08-05	1	79900
	16	15	15	2025-08-12	1	59900
	21	7	3	2025-09-04	1	74999
	25	11	1	2025-09-11	1	69900
	26	12	7	2025-09-12	1	54999

6. Find the most expensive phone.

Select * from phones order by price desc limit 1;

Result Grid

	phone_id	brand	model	price	ram_gb	storage_gb
▶	2	Apple	iPhone 16	79900	8	128

7. Find the average phone price.

Select avg(price) AS average_price from phones;

Result Grid

	average_price
▶	44579.2000

8. Count how many phones each brand has.

select brand, COUNT(*) AS total_phones
from phones
group BY brand;

Result Grid

	brand	total_phones
▶	Apple	3
	Samsung	3
	OnePlus	2
	Google	1
	Xiaomi	1
	Realme	1
	Motorola	1
	Vivo	1
	Oppo	1
	Nothing	1

9. Count how many customers are from each city.

```
select city, count (*) as total_customers
from customers
group by city;
```

	city	total_customers
▶	Kochi	5
	Thiruvananthapuram	1
	Kozhikode	1
	Thrissur	1
	Kollam	1
	Kannur	1
	Malappuram	1
	Kottayam	1
	Alappuzha	1
	Palakkad	1
	Ernakulam	1

10. Find the number of sales for each phone.

```
select phone_id, count(*) as total_sales
from sales
group by phone_id;
```

	phone_id	total_sales
▶	2	2
	4	2
	3	2
	6	3
	8	2
	10	2
	1	2
	9	2
	11	2
	5	2
	7	2
	13	2
	12	2
	14	2
	15	1

11. Find the highest and lowest phone price.

```
select max(price) as highest_price, min(price) as
lowest_price
from phones;
```

	higher_price	lower_price
▶	79900	22999

12. Find customers who made more than one purchase.

```
select customer_id, count(*) as purchase_count
from sales
group by customer_id
having count(*) > 1;
```

	customer_id	purchase_count
▶	1	2
	2	2
	3	2
	4	2
	5	2
	6	2
	7	2
	8	2
	9	2
	10	2
	11	2
	12	2
	13	2
	14	2
	15	2

13. Show all phone with their brand, model and price.

```
select phone_id, brand, model, price
from phones;
```

	phone_id	brand	model	price
▶	1	Apple	iPhone 15	69900
	2	Apple	iPhone 16	79900
	3	Samsung	Galaxy S24	74999
	4	Samsung	Galaxy A55	39999
	5	OnePlus	12R	42999
	6	OnePlus	Nord CE 4	24999
	7	Google	Pixel 8	54999
	8	Xiaomi	Redmi Note 13 Pro	25999
	9	Realme	GT 6T	30999
	10	Motorola	Edge 50 Fusion	22999
	11	Vivo	V30	33999
	12	Oppo	Reno 12	32999
	13	Nothing	Phone 2a	23999
	14	Samsung	Galaxy S23 FE	49999
	15	Apple	iPhone 14	59900

14. List customers who purchased Apple phones.

```
select distinct c.first_name, c.last_name
from customers c
join sales s On c.customer_id = s.customer_id
join phones p ON s.phone_id = p.phone_id
where p.brand = 'Apple';
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	first_name	last_name	
▶	Arun	Kumar	
	Meera	Suresh	
	Sneha	Joseph	
	Sanjay	Varma	
	Aparna	Vijay	

15. List customers who purchased Samsung phones.

```
select distinct c.first_name, c.last_name
from customers c
join sales s ON c.customer_id = s.customer_id
Join phones p ON s.phone_id = p.phone_id
where p.brand = 'Samsung';
```

Result Grid

	first_name	last_name
▶	Anjali	Nair
	Rahul	Menon
	Meera	Suresh
	Devika	Mohan
	Aparna	Vijay

16. Find each customer's total spending.

```
select c.customer_id, c.first_name, c.last_name,
       sum(s.total_amount) as total_spending
from customers c
join sales s ON c.customer_id = s.customer_id
group BY c.customer_id, c.first_name, c.last_name;
```

Result Grid

 Filter Rows:

 Export:

	customer_id	first_name	last_name	total_spending
▶	1	Arun	Kumar	134899
	2	Anjali	Nair	64998
	3	Rahul	Menon	114998
	4	Neha	Thomas	74997
	5	Vishnu	Raj	74997
	6	Akhil	Das	48998
	7	Meera	Suresh	144899
	8	Adithya	Krishnan	54998
	9	Fathima	Salim	76998
	10	Nikhil	Paul	73998
	11	Sneha	Joseph	117898
	12	Manu	Babu	87998
	13	Devika	Mohan	83998
	14	Sanjay	Varma	112899
	15	Aparna	Vijay	109899

17. Find customers whose average spending is above the overall average.

```
Select first_name, c.last_name, avg(s.total_amount) as avg_spending
from customers c
join sales s on c.customer_id = s.customer_id
group by c.customer_id, c.first_name, c.last_name
having avg(s.total_amount) > (select avg(total_amount) from sales);
```

Result Grid

 Filter Rows:

	first_name	last_name	avg_spending
▶	Arun	Kumar	67449.5000
	Rahul	Menon	57499.0000
	Meera	Suresh	72449.5000
	Sneha	Joseph	58949.0000
	Sanjay	Varma	56449.5000
	Aparna	Vijay	54949.5000

18. Find the brand with the highest total sales.

```
select p.brand, sum(s.total_amount) as total_sales
from phones p
join sales s ON p.phone_id = s.phone_id
group by p.brand
order by total_sales desc
limit 1;
```

Result Grid			Filter Rows:	
	brand	total_sales		
▶	Apple	359500		

19. List phones with more than 2 sales transactions.

```
select p.phone_id, p.brand, p.model,
       count(s.sale_id) as total_sales
from phones p
join sales s ON p.phone_id = s.phone_id
group by p.phone_id, p.brand, p.model
having count(s.sale_id) > 2;
```

Result Grid			Filter Rows:	Export
	phone_id	brand	model	total_sales
▶	6	OnePlus	Nord CE 4	3

20. Show customers and the phone model they purchased.

```
select c.first_name, c.last_name, p.brand, p.model
from customers c
join sales s on c.customer_id = s.customer_id
join phones p on s.phone_id = p.phone_id;
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Cont
	first_name	last_name	brand	model	
▶	Sneha	Joseph	Apple	iPhone 15	
	Meera	Suresh	Apple	iPhone 15	
	Sanjay	Varma	Apple	iPhone 16	
	Arun	Kumar	Apple	iPhone 16	
	Meera	Suresh	Samsung	Galaxy S24	
	Rahul	Menon	Samsung	Galaxy S24	
	Rahul	Menon	Samsung	Galaxy A55	
	Anjali	Nair	Samsung	Galaxy A55	
	Fathima	Salim	OnePlus	12R	
	Nikhil	Paul	OnePlus	12R	
	Anjali	Nair	OnePlus	Nord CE 4	
	Neha	Thomas	OnePlus	Nord CE 4	
	Neha	Thomas	OnePlus	Nord CE 4	
	Manu	Babu	Google	Pixel 8	
	Arun	Kumar	Google	Pixel 8	
	Akhil	Das	Xiaomi	Redmi Not...	
	Vishnu	Raj	Xiaomi	Redmi Not...	